

$$M: X_i \sim N(\alpha + \beta \cdot t_i, \sigma^2)$$

a)

$$\begin{aligned} \sum_{i=1}^n t_i - \bar{t} &= n \cdot \bar{t} - \sum_{i=1}^n t_i = n \cdot \frac{1}{n} \sum_{i=1}^n t_i - \sum_{i=1}^n t_i = 0 \\ SPD_{XT} &= \sum_{i=1}^n (X_i - \bar{X})(t_i - \bar{t}) = \sum_{i=1}^n X_i(t_i - \bar{t}) - \sum_{i=1}^n \bar{X}(t_i - \bar{t}) \\ &= \sum_{i=1}^n X_i(t_i - \bar{t}) - \bar{X} \cdot \sum_{i=1}^n (t_i - \bar{t}) = \sum_{i=1}^n X_i(t_i - \bar{t}) \end{aligned}$$

b)

"linearkombinationer af normale variable er normale da skal vi bare finde middelværdi og varians

$$\begin{aligned} \mathbb{E}[SPD_{Xt}] &= \mathbb{E} \left[\sum_{i=1}^n X_i(t_i - \bar{t}) \right] = \sum_{i=1}^n \mathbb{E}[X_i](t_i - \bar{t}) \\ &= \sum_{i=1}^n (\alpha + \beta \cdot t_i)(t_i - \bar{t}) = \alpha \cdot \sum_{i=1}^n (t_i - \bar{t}) + \sum_{i=1}^n \beta t_i (t_i - \bar{t}) \\ &= 0 + \beta \cdot \sum_{i=1}^n t_i (t_i - \bar{t}) - \beta \cdot \sum_{i=1}^n \bar{t} (t_i - \bar{t}) = \beta \cdot \sum_{i=1}^n (t_i - \bar{t})^2 = \beta \cdot SSD_t \end{aligned}$$

Noter :

$$\begin{aligned} \sum_{i=1}^n \bar{t} (t_i - \bar{t}) &= \bar{t} \sum_{i=1}^n (t_i - \bar{t}) = 0 \\ \mathbb{V}[SPD_{Xt}] &= \mathbb{V} \left[\sum_{i=1}^n X_i(t_i - \bar{t}) \right] = \sum_{i=1}^n \mathbb{V}[X_i](t_i - \bar{t})^2 = \sigma^2 \cdot SSD_t \end{aligned}$$

Andet lighedstegn kommer fra at X_i 'erne er uafh per antagelse

Da

$$SPD_{Xt} \sim N(\beta \cdot SSD_t, \sigma^2 \cdot SSD_t)$$

Vi ved hvordan estimatet for hældningen ser ud

$$\hat{\beta}(X) = \frac{SPD_{Xt}}{SSD_t}$$

Igen normalfordelt vi finder parametre

$$\begin{aligned} \mathbb{E}[\hat{\beta}(X)] &= \frac{\mathbb{E}[SPD_{Xt}]}{SSD_t} = \frac{\beta \cdot SSD_t}{SSD_t} = \beta \\ \mathbb{V}[\hat{\beta}(X)] &= \frac{1}{SSD_t^2} \cdot \mathbb{V}[SPD_{Xt}] = \frac{\sigma^2}{SSD_t} \end{aligned}$$

Da

$$\hat{\beta}(X) \sim N\left(\beta, \frac{\sigma^2}{SSD_t}\right)$$